

**Points, Paychecks, and Performance:
Exploring the Financial Architecture of NBA
Roster Construction Strategies**

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Abstract

This paper explores the multifaceted relationship between NBA player statistics, contract valuation, and the league's evolving salary cap structure. Drawing from a wide range of analytical models and scholarly articles, the research examines which performance metrics, ranging from basic counting stats like points per game to advanced metrics like PER and true shooting percentage, most significantly influence contract size. Additionally, this paper evaluates how the 2023 NBA Collective Bargaining Agreement (CBA) has reshaped cap management strategies, with new rules regarding punitive thresholds like the First and Second Aprons. By synthesizing findings from both academic and industry sources, the paper highlights how teams now balance statistical evaluation with financial constraints to optimize roster construction and player compensation. Ultimately, the research underscores the NBA's shift toward more data-informed decision-making while promoting equitable pay structures under the revised CBA.

Keywords: Salary Cap Efficiency, Player Evaluation, Roster Construction, Collective Bargaining Agreement

Introduction

Before the 2011–12 NBA season, the league entered a lockout while negotiating a new Collective Bargaining Agreement (CBA) that would provide fair terms for both players and teams. The resulting agreement, implemented in the winter of 2011, marked a turning point in how NBA rosters were constructed. Since then, each successive CBA has added new layers of complexity to team-building decisions, culminating in the 2023 CBA – a major topic of discussion in the NBA community due to its stricter financial penalties and expanded salary cap restrictions.

The presence of the CBA has significantly influenced how teams evaluate players, develop playstyles, and balance offensive versus defensive priorities. With heightened consequences for inefficient spending under the 2023 CBA, front offices are increasingly focused on optimizing roster construction and allocating salary in the most strategic way possible.

This paper seeks to understand which types of players are rewarded with larger shares of team salary and which team-level strategies most effectively lead to regular-season success. In other words, how are NBA teams valuing talent in the modern landscape, and what styles of play translate to winning?

To answer these questions, three models were developed. The first examines how individual characteristics, such as position, age, and performance metrics, affect a player's share of team salary. The second analyzes which team playstyle indicators are most strongly associated with win percentage. The third compares the relative impact of offensive versus defensive performance on team success.

These findings aim to provide insight into current roster construction strategies and may help both industry professionals and fans better understand the implications of the CBA and the metrics that contribute most to team performance.

Literature Review

To understand the research I have completed for this literature review most efficiently, I have formatted the following review into six sections: (1) an introduction to salary cap, (2) the presence of competitive balance, (3) the tension between winning and profit-maximizing, (4) the fundamental makeup of successful teams, (5) the use of modeling to identify player values, (6) and the implementation of the 2023 Collective Bargaining Agreement. The primary objective of these sections is to explore optimal strategies for cap management and salary allocation for NBA teams, with the goal of enhancing team performance while also considering the franchise's financial objectives.

Introduction to Salary Cap

Sports is a multi-billion-dollar industry, and there are no signs of it slowing down, but how much of that revenue is disproportionately made by the most popular teams in the Big Four American Sports Leagues – NBA, NFL, MLB, and NHL? Due to the most popular teams having an unfair advantage by possessing more salary-related funds from heightened revenues, leagues can sustain less parity. To combat this, a salary cap system can be implemented in a given league to put a limit on roster-related spending. The addition of a salary cap system to a league may be tricky due to it potentially limiting player earnings; however, the salary cap can be structured around a “share of defined revenues” (Zimbalist, 2010). Thus, the players will still be given a fair share of the league revenue. Opening the door to the use of a salary cap system allows for a league to promote parity and other ideologies that may boost league performance, which serves the best interests of the league. Now, the addition of a salary cap could have a negative impact on league revenue because the most popular franchises can’t simply pay for all of the stars, which in turn limits the potential profits for those franchises. However, there are findings that imply an increase in competitive balance and reasonable profits for large or small market franchises can be had if such a system is in place (Késenne, 2000). Consequently, discussion around a salary cap commands a discussion around competitive balance and its importance to professional sports.

Competitive Balance

Parity in professional sports is a necessity, but how does a salary cap comply with the idea of parity or competitive balance. Having a system in place is a good start, however an analysis of the competitive balance with the cap system still needs to be done. Research was done across multiple professional sports leagues regarding this topic and researchers found that the NBA had the least parity out of all the big American sports leagues with a salary cap because the teams who spent more were still more likely to have greater success (Shorin, 2017). Now, the authors concluded these findings in 2017, way before the current CBA within the NBA. Nevertheless, the findings force the notion of competitive balance being an issue within the NBA, despite a complex salary cap system being in place. Another article presented the findings that a cap system may even decrease competitive balance, while suggesting that athletes act in their own self interest (Totty, 2011). Thus, if athletes act in their own self interest then how might franchises act while deciding who to pay. There are two groups of teams within professional sports regarding thought process during roster construction – those who want to maximize their team success and those who want to maximize their profits.

Winning and Profit-Maximizing

Each decision during roster construction in the NBA has great implications on the rest of the given roster and on the rest of the league, so while teams make these decisions they are forced to make them based upon a win-maximizing ideology or a profit-maximizing ideology. The authors of one article had the goal of understanding how and why different NBA franchises

operate between 2001 and 2019, and they found that these two groups can very much be identified (Chatzistamoulou, 2021). They also suggested that a salary cap system can push teams to manage their allocation of player salaries more efficiently to achieve a combination of team and economic success. This idea of ‘efficiency’ continues throughout many articles that were published regarding the decision of either win-maximizing or profit-maximizing, which leads to another article that describes the process of identifying the franchises that make player salary related decisions in the most efficient manner possible. A payroll efficiency metric was created to locate teams that “win more with less”, which generally were found to be small market teams due to them having more money constraints (Scott, 2020). The discussion on payroll efficiency disparity within the NBA and why that might be continues to resonate with the idea that certain teams have to maximize either their profits or team success and that decision is reflected in a given team’s recent body of work. An example of this type of decision could be the idea of the “Star Strategy”, which is the strategy of crafting a star player to increase profits and fan engagement (Guan, 2022). This strategy suggests that teams might be willing to overspend some of their salary cap to acquire or retain their star player in order to maximize their profits, thus being a profit-maximizing strategy.

On the other hand, a strategy that may be viewed as a win-maximizing strategy is detailed by Martin B. Schmidt, where he describes an economical approach to team building in the NBA to maximize success. In his findings, he suggests that team productivity increases if a firm enables its workers to focus on their strengths and excel in the areas they perform best (Schmidt, 2020). Relative to the NBA, these findings harmonize with the strategy of constructing a balanced roster that has players with differing talents, facilitating a productive team environment and maximizing team success. The final paper that includes the win-maximizing belief is from Rory Robinson and details what NBA championship rosters have in common. One trait that was found to be common amongst championship-caliber teams was that the star player wasn’t paid more than 25% of the team’s total salary, which can be described as a win-maximizing decision rather than a profit-maximizing one (Robinson, 2022). This final article pertaining to win and profit-maximizing encouraged a deeper look into how successful teams are made. Though the above trait can be used to maximize success, it may also be looked at as a fundamental building block for the blueprint to be successful in the NBA.

Fundamental Makeup of Successful Teams

If we assume that a team wants to maximize success then they may look towards what brought others success in the past. This ideology has earned the NBA the title as a ‘copy-cat league’, meaning that teams will build their roster to be successful in the same way as others were previously. Thus, many discussions surrounding the foundation of a winning team or how to construct a championship roster began. The authors for the first article referenced in this section conducted an analysis of the concentration of payroll in the MLB and NBA, which is how they concluded that the two sports had opposite optimal strategies for salary allocation amongst players (Chen, 2023). Their findings suggest that teams in the MLB may want to spread

their salary across multiple great players and not just one star, but it is important to note that once again the MLB does not have a salary cap system like the NBA so inferences may be hard to draw from the findings. Nevertheless, the authors also claimed that teams in the NBA may want to pay for a superstar in order to maximize their success.

Following the article, I found another that consisted of greater detailed analysis on NBA teams' salary allocation and their efficiency. The author of this article, Tariq Abubakr, created an efficiency metric to analyze how teams were allocating their salary cap space, specifically if it is more efficient regarding team success to pay one or two stars or many great players. Abubakr found that teams who distribute their cap space to multiple great players have a better chance of making the playoffs when compared to a team with a couple high-salary stars, but it was the teams with the couple high-salary stars that had greater success in the playoffs (Abubakr, 2021). These findings make sense due to the demanding 82-game regular season, contrasted with a playoff, where teams typically rely on a rotation of approximately eight players per game. Another way in which teams may look to copy another team's winning formula is by analyzing what the players on a successful team actually contributed to that team. This question was asked by Sameer Deshpande in another article where he had the aim of understanding what winning players had in common, contrary to just analyzing the team as a whole (Deshpande, 2016). An example of this analysis may be finding players on successful teams that excelled in one aspect of the game, and if that type of player was common across successful teams then that type of player could be viewed as a necessity, like a defensive minded player who could substitute as a scoring option from time to time such as Derrick White or Jrue Holiday. Before continuing with more salary cap allocation strategies, it might be important to note a paper from my research that described the "Three Pillars Approach". The idea of this approach is that teams must create an identity before they have success, with the author including multiple examples of teams that follow this identity guideline (Spins, 2022). The concept of having an identity is important because it allows a group of players to mesh well with each other, indicating that teams may incorporate this concept into their roster construction.

Now, the author of an article that analyzes pay and performance in the NBA, Kevin J. Stiroh, set out to uncover the reality of the "Contract Year" fallacy. This describes the idea that athletes will perform to a higher level when they are in the final year of their current contract and want to sign a lucrative new one. The findings from the article suggest that the fallacy may be true, and players perform to the best of their abilities in that season (Stiroh, 2007). These findings may be utilized to understand certain cap strategies for NBA teams if they wish to maximize their team's success.

Modeling to Identify Player Values

The process of identifying potential players for a team is inherently time-consuming, but negotiating and agreeing upon the right amount of money relative to the value that player brings to a team adds an additional layer of complexity to the process. In my research, I found three articles with the same goal in mind: identify which statistics are most valuable when earning a

contract in the NBA. The authors of the first article found that simple counting stats, like points, and high volume will lead to bigger contracts (Sigler, 2018). Intuitively, this was the easiest conclusion to agree with because players that play more will almost always have more stats, which leads to more appealing statistics. Next, Kenyan Wei brought an updated prospective into the mix, which was that field goal percentage and plus minus were the most important metrics for earning a large contract (Wei, 2023). This conclusion suggests that NBA teams don't simply value simple counting stats, but instead may look towards more advanced efficiency metrics while negotiating contracts. This inference became increasingly likely when I got to the third article relating to this topic, where the author once again created a predictive model to analyze what statistics are most valuable when earning a contract in the NBA. Daniel Wang highlighted Value Over Replacement Player (VORP) and Points Per Game (PPG) as having a positive relationship with earning a bigger contract within his research, where VORP represents advanced statistics, and PPG represents simple counting stats (Wang, 2024). Overall, these three articles suggest that varying advanced and basic metrics may have a big impact on the size of contracts within the NBA, however they can be merged to show how NBA teams use a combination of statistics to analyze player value, and the process can't simply be broken down into a few variables.

Another author, Andrew Weiner, decided to create a linear regression model to predict salaries based on varying statistics, which is similar to the previous approaches but views the player value process from a different perspective. The model that was used was able to accurately predict salaries for middle of the pack players, however it resulted in underpaying elite players or overcompensating low-quality players (Weiner, 2023). Now, analyzing player value relative to salary is crucial, but simply modeling for player value alone can also bring key insights to the discussion. Muniz and Flamand are two authors that had the goal of understanding how to build the most balanced team in order to maximize team success. This aim resides with the ideology of win-maximizing that was previously mentioned, however the implementation of models to predict player performance and player values allowed the authors to make meaningful conclusions pertaining to the aim (Muniz, 2022). Similarly, Nick Kalinowski created a model using LEBRON, which is an all-encompassing metric that can be used to evaluate a player's overall performance. The author used this metric to analyze how teams may value players, specifically rookies, where he used the 2023 NBA draft class to analyze the model's accuracy (Kalinowski, 2024). Having metrics that can assign values to a player's overall performance can be insightful, but what if teams were to analyze the offensive and defensive sides of the ball separately. Jim Lackritz, the co-founder of the Sports MBA program at San Diego State University, created a point scheme to determine a player's contributions to a team's offensive performance. Through this scheme, conclusions can be drawn regarding a player's offensive value in relation to the team's offensive success, though no such point scheme was created for defensive contributions (Lackritz, 2020). The model created within the article above can be helpful in identifying players' offensive value, but that's only half the game.

The final article in this section was created, in part, by two Syracuse University professors – Justin Ehrlich and Shane Sanders- along with Christopher Boudreaux. They analyzed discriminations in pay between offensive and defensive talents in the NBA, finding that players on their first contract were more likely to be paid for their offensive talents, while veterans and players who were on their second, third, or even fourth contract were more likely to be paid for their defensive talents (Ehrlich, 2019). The analysis was conducted with the use of offensive adjusted plus minus and defensive adjusted plus minus, which are stats that can define a player's offensive or defensive play in a single statistic. Modeling can be very useful when identifying player values and how much a team should in fact pay for a certain player, but simultaneously a team must be aware of the constraints of the salary cap system when negotiating a new contract, which is where the 2023 Collective Bargaining Agreement comes into effect.

2023 Collective Bargaining Agreement

NBA contracts have many nuances that can offer teams flexibility while maneuvering the salary cap system, some of these nuances were described within an article on The Hoop Blog. In this article the author describes different contract conditions, like guaranteeing a salary or implementing incentives or bonuses. When teams can negotiate a non-guaranteed contract it offers flexibility within managing their payroll and might even cause players to play to a higher level to earn the guaranteed money, however players now will often secure a guaranteed contract due to a heightened risk of injury. Incentives and bonuses may also have a positive effect on player performance, suggesting that players will perform if they have the potential to earn more (TheHoopBlog, 2023). In another article, Jeffrey Goldrosen analyzed the NBA's salary cap structure in 2018 and how it impacted players and organizations. The author's findings suggest that the system, at the time, was more beneficial for owners than players (Goldrosen, 2018). These conclusions come as no surprise to me due to earlier articles indicating that a salary cap may reduce potential player salaries, therefore limiting team spending on player payroll. Albeit the NBA and the National Basketball Players Association decided to work tirelessly together to create the CBA in 2023 so that it would benefit any and all players in the league. I believe that Goldrosen's findings are accurate, which is why so many changes were made to the salary cap system in 2023.

Bryan Toporek, a writer for BleacherReport, put out an article that described a ton of changes that were made to the system via the 2023 CBA. A few details that he highlighted were a look into longer extend-and-trade contract lengths from a maximum of 3 years to 4 years with extension limits raised from 120% of a player's previous salary to now 140% of a player's previous salary and forcing teams to reach the cap floor before the first day of the regular season (Toporek, 2024b). In previous years, teams would have to reach the floor by the end of the regular season, meaning that they had to spend 90% of the salary cap by that time. This change meant that more players would be earning contracts sooner rather than later in order for teams to meet the cap floor requirements. The author also introduced the First and Second Aprons, which

can be viewed as loose limits on total salary which allow a team to exceed the margins but provides consequences for those teams. Another article, from Luke Adams, analyzed rules associated with the First and Second Aprons and what happens when teams become “Hard-Capped”. Becoming hard-capped simply means that a team is utilizing a roster construction tool that is removed for teams over the First or Second Apron, therefore if a team wants to acquire a player via sign-and-trade, for example, then they will become hard-capped at the First Apron due to the sign-and-trade tool being striped from teams over this limit (Adams, 2024). The author goes on to discuss many more potential ways in which teams can become hard-capped, which will be useful amongst all discussions relating to the 2023 CBA and how teams maneuver through roster construction. An additional article also analyzed the CBA, in it the author noted rule changes, like easing and tightening of tampering rules (contradictory but correct), increased potential payout and lengths for 2nd round rookies, and the tightening of underspending teams (Vorkunov, 2024). These changes suggest that, as I stated before, the NBPA and the NBA continue to work together to create rules that would benefit the association and its players, which are made evident in the rule changes that supply less talented players with the opportunity to earn more money.

The final article that I analyzed regarding the CBA detailed the new TV deal that the NBA has negotiated, which will bring large amounts of revenue to the league, directly causing the salary cap to feel upwards pressure. Once again, this article was written by Bryan Toporek, in it he states that the CBA will only allow salary cap to increase by at most 10% annually and will never decrease year to year (Toporek, 2024a). The capped salary increase enables a controlled rise in player salaries rather than forcing teams to pay one offseason of free agents enormous amounts of money because the TV deal kicked in. Again, these changes within the CBA show an effort from the league and its player association to ensure that every player is treated equitably and afforded equal opportunities. Further research related to the CBA may prove useful in analyzing cap management strategies for 2024 and beyond.

Data Summary

This paper uses publicly available NBA data from Basketball-Reference.com, which is a comprehensive database that contains team and player statistics spanning the history of the league. This dataset includes traditional counting statistics (e.g., points per game, assists per game), advanced metrics (such as Player Efficiency Rating and True Shooting Percentage), and team-level performance indicators like win percentage and offensive/defensive rating.

Additionally, NBA salary data was obtained from Basketball-Reference.com and Basketball.RealGM.com. The former provides annual team payrolls and individual player salaries, while RealGM offers season-by-season league-wide financial benchmarks, including the salary cap, luxury tax threshold, first apron, and, beginning with the 2023 Collective Bargaining Agreement, the second apron. These datasets were combined to calculate each team’s spending as a percentage of the league salary cap and relative to other teams in a given season.

To ensure that the analysis in this paper focuses on meaningful contract decisions, the sample was limited to players who participated in at least the average number of games and the average number of minutes per game during the sample period: 41 games and 18.9 minutes per game in a given season. These criteria were designed to exclude end-of-bench players and short-term injury replacements, ensuring that the sample represents players typically included in a team's active rotation - the top nine players or so - whose salaries reflect intentional roster-building decisions. These filters yielded a dataset comprising 3,176 NBA player-seasons across 420 team rosters.

Together, these datasets enable analysis of on-court success and financial decisions regarding roster construction across multiple seasons. All data was collected from the 2011–12 season through the 2024–25 season. This time frame was selected because the 2011 NBA lockout introduced the concept of the ‘apron,’ which imposed additional consequences on high-spending or inefficient teams, leading to greater planning and preparation on how teams should spend their money on players.

Summary Statistics

Before visualizing the data, it is helpful first to organize key metrics into descriptive statistics. Tables 1 and 2 present player season-level variables, including both traditional counting statistics – such as Points per Game, Rebounds per Game, and Assists per Game – and advanced efficiency metrics, such as Player Efficiency Rating (PER), True Shooting %, and Win Shares per 48 minutes. Table 3 highlights team-level variables that characterize playstyle or team archetype, such as Team 3-Point Attempt Rate and many Offensive and Defensive Four Factors. Additionally, in Table 4 there are variables related to contracts and salaries, such as Percentage of Team Salary, which indicates the proportion of total team salary that a player occupies for a given season, and Percentage of Salary Cap, which represents the percentage of team salary relative to the salary cap for that season. Lastly, Team Win%, Offensive Rating relative to league average for a given season, and Defensive Rating relative to league average for a given season are included as measures of team success, providing a baseline for evaluating effective roster construction strategies.

Table 1: Player Performance Descriptive Statistics

	Age	Games (G)	Minutes per Game (MPG)	3P%	3 Point Attempts per Game	Rebounds per Game (RPG)	Assists per Game (APG)	Assists:Turnover	Steals per Game (SPG)	Blocks per Game (BPG)
Min.	19.00	41.00	18.90	0.00	0.00	1.20	0.10	0.08	0.10	0.00
1st Quart.	23.00	58.00	23.40	30.80	1.80	3.10	1.40	1.25	0.60	0.20
Median	26.00	68.00	27.90	35.10	3.50	4.30	2.20	1.67	0.80	0.40
Mean	26.42	66.17	27.82	31.78	3.46	4.97	2.89	1.78	0.89	0.54
3rd Quart.	29.00	76.00	32.12	38.30	4.90	6.30	3.90	2.21	1.10	0.70
Max.	40	82	39.4	100	13.2	16.00	11.7	7.33	3.00	3.80
Count (N)	3176	3176	3176	3176	3176	3176	3176	3176	3176	3176

Table 2: Player Performance Descriptive Statistics (cont.)

	Points per Game (PPG)	Player Efficiency Rating (PER)	True Shooting %	Usage %	Win Shares per 48 (WS/48)	Box Plus Minus (BPM)	Offensive Box Plus Minus (OBPM)	Defensive Box Plus Minus (DBPM)
Min.	2.90	3.70	41.30	6.50	-4.03	-6.60	-5.50	-3.30
1st Quart.	8.80	12.20	52.90	16.20	3.26	-1.40	-1.30	-0.80
Median	11.80	14.80	55.80	19.50	4.85	0.00	0.00	0.00
Mean	12.99	15.38	56.02	20.15	5.06	0.32	0.29	0.02
3rd Quart.	16.20	17.90	59.00	23.60	6.64	1.70	1.50	0.70
Max.	36.10	32.80	74.50	41.70	15.46	13.70	10.30	4.60
Count (N)	3176	3176	3176	3176	3176	3176	3176	3176

As previously described, Tables 1 and 2 show summary statistics for variables that may be used in a potential Player Evaluation Model to explain what metrics may be most important to earning a larger share of total team salary. The aforementioned filters regarding minimum play time requirements can be seen within the statistics for Games Played (G) and Minutes per Game (MPG), with 41 games and 18.9 minutes as the minimum values. A large variance within many of the metrics can be seen as well, specifically in advanced metrics, such as Box Plus Minus (BPM) and Player Efficiency Rating (PER). Additionally, Assists per Game (APG), Points per Game (PPG), and Rebounds per Game (RPG) see a big jump from the 3rd quartile to their max values within the dataset, suggesting that elite players who perform very well in those categories will earn a greater contract and for good reason. It is also interesting to note that 50% of the player sample is between 23 and 29 years old, which is around the time players are entering their prime or reaching their absolute peak athletically. This can also be viewed as the years in the NBA following the end of a player's rookie-scale contract, in which they are either signed in free agency by a new team or re-signed/extended by their current team. Intuitively, it makes sense that teams would like to roster as many players in their primes as possible to utilize every player's full potential, therefore maximizing the team's potential. One final metric worth highlighting is 3-Point Percentage (3P%) due to its minimum and maximum values being 0% and 100%, respectively. These outliers are caused by players with few attempts over the course

of a season, but in those attempts, they either make or miss all of them. To account for this, 3-Point Attempts per Game will be included in any Player Evaluation Model that 3P% is included in.

Table 3: Team Playstyle Descriptive Statistics

	Adjusted Team 3PA Rate	Adjusted Team 3P%	Adjusted Team FTA Rate	Adjusted Team eFG%	Adjusted Opponent eFG%	Adjusted Team TOV%	Adjusted Opponent TOV%	Team ORB%	Team DRB%
Min.	-9.54	-5.26	-7.39	-4.83	-4.35	-2.88	-3.07	17.90	69.10
1st Quart.	-2.79	-1.07	-1.57	-1.34	-1.00	-0.63	-0.67	22.10	74.47
Median	-0.39	-0.09	-0.18	0.08	0.00	0.06	-0.04	23.95	76.00
Mean	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.11	75.86
3rd Quart.	2.60	1.09	1.54	1.23	1.08	0.56	0.68	25.90	77.33
Max.	16.50	6.32	10.15	6.05	4.10	2.70	3.48	32.60	81.60
Count (N)	420	420	420	420	420	420	420	420	420

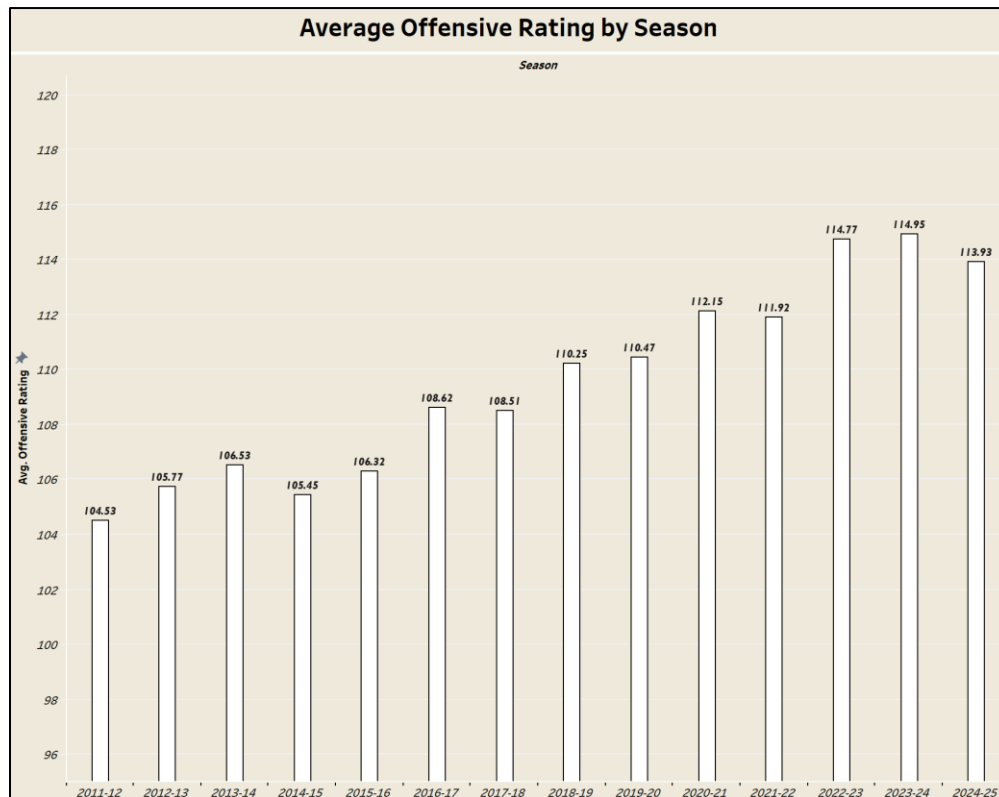
Before analyzing these statistics, it is important to note that for each metric labeled as “adjusted,” the values have been normalized relative to the league average for the corresponding season. For example, given the NBA’s ever-evolving nature, particularly the substantial increase in three-point attempts over time, each team’s three-point attempt rate has been adjusted to reflect its deviation from the league average in the given season in which the team competed. Now, the first thing that jumps out when looking at Table 3 is the spread for both 3-Point Attempt Rate and 3-Point Percentage, highlighting how differently teams approach shot selection within the NBA and how successful, or unsuccessful, teams are at adapting to the “3-Point Evolution”. Adjacent to 3-Point Attempt Rate is Free Throw Attempt Rate, which also contains a large spread between the minimum and maximum values, punctuating the differing team playstyles, particularly in how aggressively teams attack the rim or generate offense through drives and contact. These metrics can be useful when compared to a team’s success through metrics, such as Win Percentage, which can be seen in Table 4.

Table 4: Team Success and Salary Figures Descriptive Statistics

	Team Win%	Offensive Rating vs League Average	Defensive Rating vs League Average	Percentage of Team Salary	Percentage of Salary Cap
Min.	10.61	-9.95	-7.01	0.23	72.97
1st Quart.	39.02	-1.97	-1.88	2.91	105.20
Median	51.22	0.23	0.06	6.56	116.89
Mean	49.96	0.16	0.11	8.92	116.12
3rd Quart.	60.98	2.39	2.04	13.19	122.98
Max.	89.02	8.25	7.90	35.48	174.83
Count (N)	420	420	420	3176	420

Similar to the previous table, certain metrics, like Offensive Rating relative to the league average and Defensive Rating relative to the league average, are adjusted based on the season in which the team competed. These metrics are considered to be better than raw Offensive and Defensive Ratings because the NBA has seen an increase in scoring driven by a rise in three-point attempts, which means these ratings have naturally risen for all teams, forcing analysis to account for the year. For context, the average Offensive Rating in the NBA during the 2011-12 season was 104.53 and increased to 114.95 by the 2023–24 season, this trend can be seen in Figure 1. Like prior metrics, most of the variables within Table 4 have a compact 1st-3rd quartile spread, but a large range between minimum and maximum values. This emphasizes the idea that there may have been a parity issue within the league during the time of this sample, strengthening the thought that the 2023 Collective Bargaining Agreement employed many new techniques to increase competitiveness across the league. Additionally, it is evident that the salary cap is a very loose limit for team spending, since at least 75% of the sample spent more than the salary cap for their respective season. Furthering the need for extra limits and aprons, as introduced in 2023, to combat teams being able to simply pay more to win more.

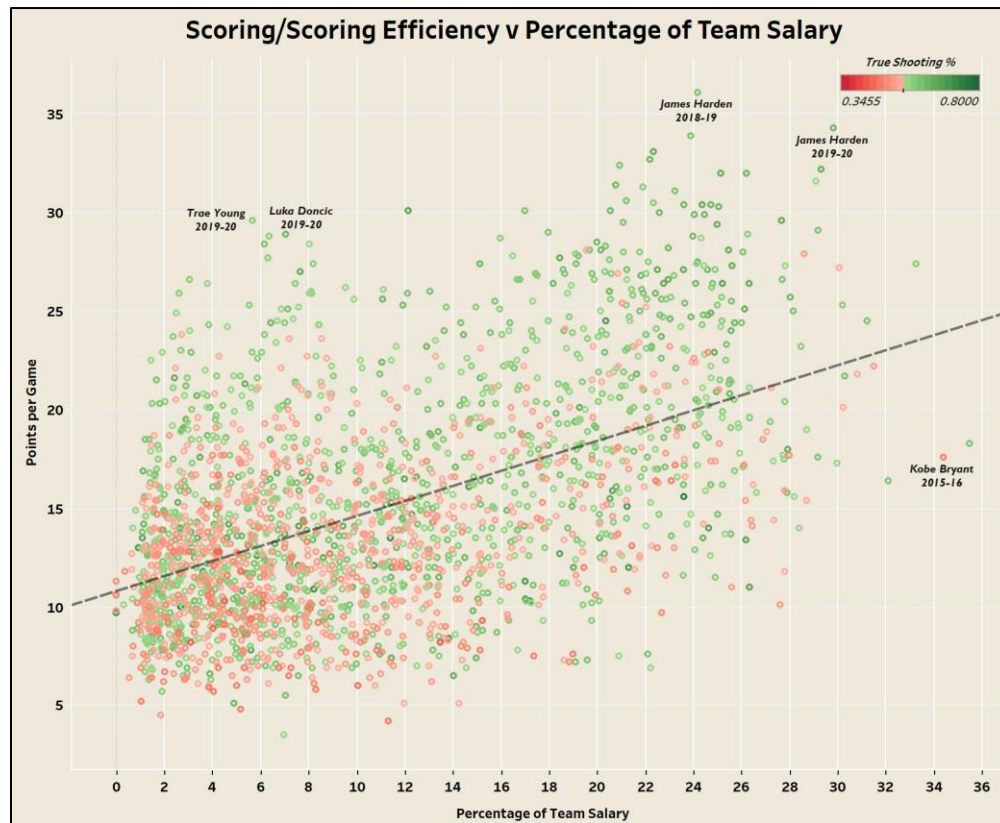
Figure 1: Average Offensive Rating by Season (2011-2025)



Pre-Model Visualizations

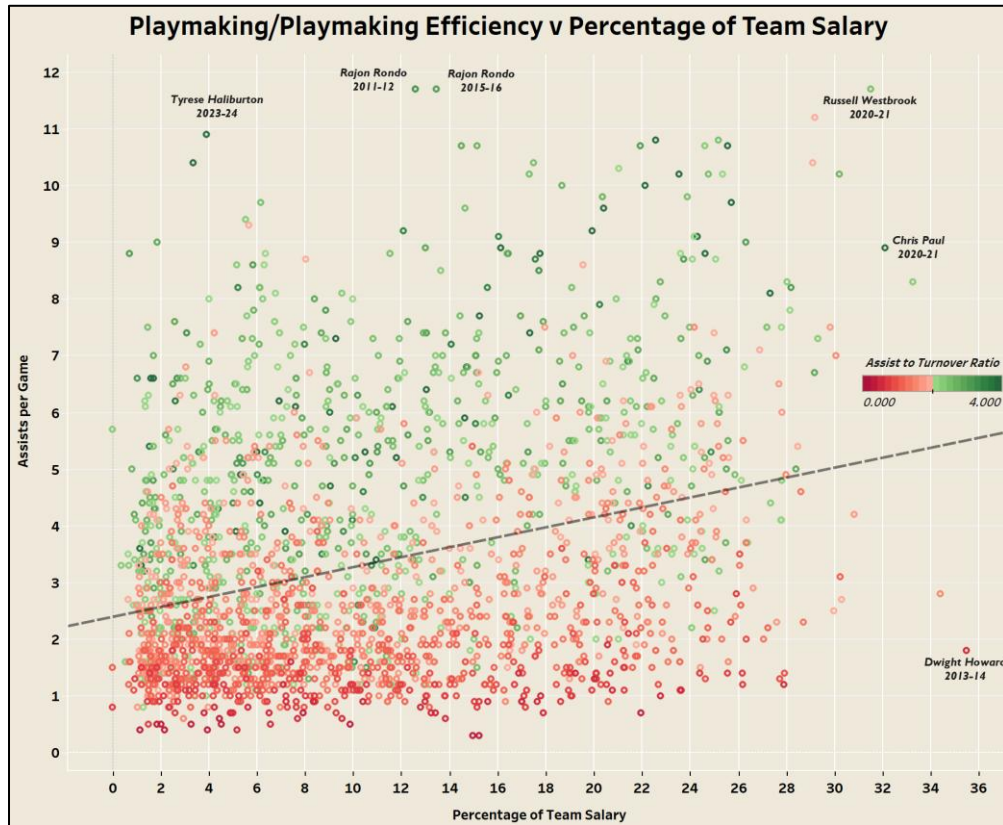
Now, visualizing the data may help produce an understanding of how the independent variables may affect dependent variables in future models, like contract size or team success. The visualizations will provide insight into relationships before regression models are created and analyzed. The first of many graphs, Figure 2, plots the relationship between Points per Game and Percentage of Team Salary.

Figure 2: Scoring/Scoring Efficiency vs Percentage of Team Salary Taken Up by a Given Player



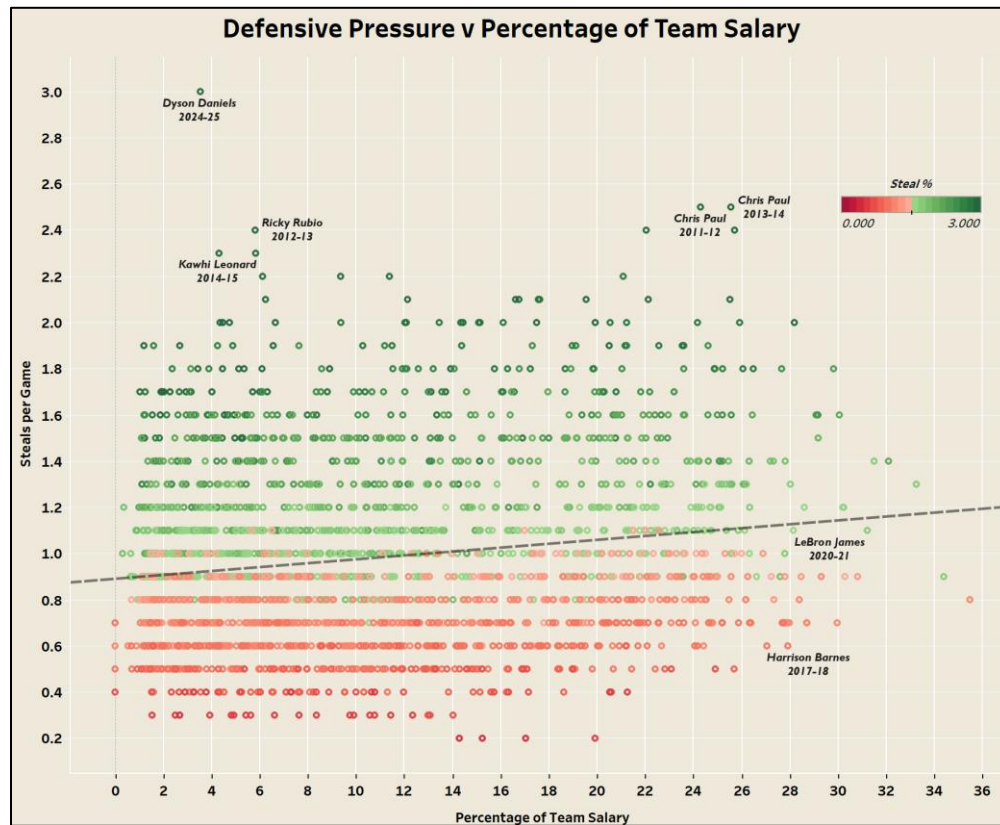
As seen in Figure 2, Points per Game has a strong positive correlation with the percentage of team salary taken up by a given player, or in other words, the size of a player's contract in relation to the team's total salary. This suggests that players who score more will most likely earn a greater salary. Some examples of this are labeled on the graph, including prime James Harden in the top right corner, when he was earning 25%-30% of the Houston Rockets' total team salary but providing high levels of scoring with good efficiency. A few notable examples of players significantly outperforming their expected Points per Game relative to their salaries include Luka Dončić and Trae Young during their second NBA seasons. This highlights the importance of teams hitting on their rookie picks/maximizing the value from their rookie-scale contract players, which is a common theme throughout the rest of the graphs. Also, it's important to note that a lot of times players will underperform and not reach the level of play expected from them, given the size of their contract, one example being Kobe Bryant during his final season in the NBA.

Figure 3: Playmaking/Playmaking Efficiency vs Percentage of Team Salary Taken Up by a Given Player



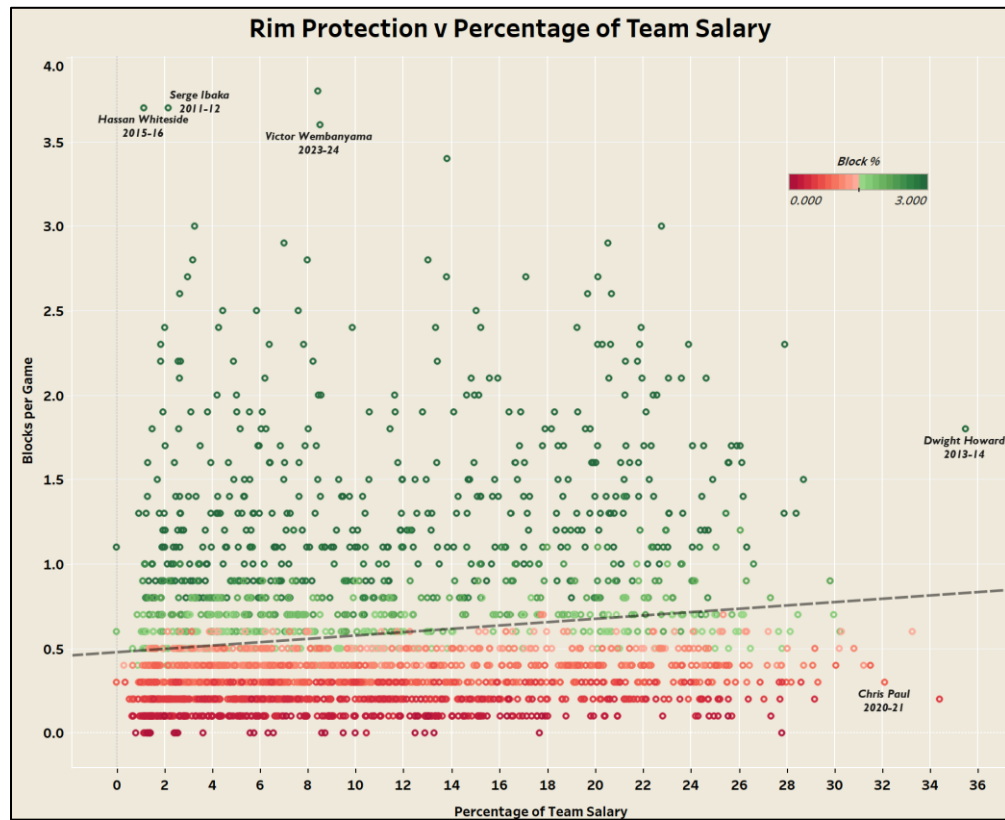
Continuing with offensive impact in Figure 3, the relationship between playmaking and salary relative to the team's total salary is also positive but slightly weaker than scoring's relationship with salary. A couple of noteworthy names that outperformed their expected Assists per Game numbers while playing on relatively small contracts are Tyrese Haliburton and Rajon Rondo. Haliburton was in the final year of his rookie-scale contract when he averaged almost 11.0 APG, while Rondo was known to be an affordable pass-first point guard during the middle years of his career. Russell Westbrook and Chris Paul, towards the tail-end of their primes, were also elite playmakers who outperformed their expected playmaking numbers but were being paid comfortably for it. Now, an example of a player who underperformed as a playmaker was Dwight Howard at the tail-end of his prime, however, unlike Kobe this makes sense, and teams might be okay with this level of output because players with Howard's skillset are expected to "earn" their contract through scoring and defense rather than playmaking. Again, the importance of maximizing players on rookie-scale contracts is seen within these visualizations.

Figure 4: Defensive Pressure (Steals per Game) vs Percentage of Team Salary Taken Up by a Given Player



In Figure 4, the relationship between defensive pressure and salary is on display, and while it's not as strong as the previous offensive metrics' relationships, it is still positive. Chris Paul recorded multiple seasons in which he averaged about 2.5 steals per game—an elite level of defensive output relative to his salary. These performances represent a significant overproduction for a point guard occupying roughly 25% of his team's total salary cap, and are comparable in value to James Harden's high-scoring seasons previously discussed. Similarly, several players on rookie-scale contracts—such as Kawhi Leonard, Ricky Rubio, and Dyson Daniels—appear in the top-left quadrant of Figure 4, indicating that they exceeded expectations in Steals per Game despite their lower salaries, emphasizing the importance of successful drafting. Conversely, instances of underperformance relative to salary are also evident. For example, Harrison Barnes during the 2017–18 season received a substantial portion of the Dallas Mavericks' payroll though he averaged around .5 Steals per Game. Finally, LeBron James in the 2020–21 season serves as a case in which a highly compensated player delivered performance commensurate with his salary, particularly as a point-of-attack defender.

Figure 5: Rim Protection (Blocks per Game) vs Percentage of Team Salary Taken Up by a Given Player



Taking a look at another metric for defenders in the NBA, Figure 5 shows the relationship between Blocks per Game and player salary. Similar to the previous relationship between Steals per Game and player salary, the correlation in the graph above is weakly positive. Once again, players that outperform their rookie-scale or minimum contracts are seen in the top left – Whiteside, Ibaka, and Wembanyama. Simultaneously, many players underperform as rim protectors, considering their salary, such as undersized guards like Chris Paul, who is identified in the above figure. Previously, Dwight Howard was recognized as a poor playmaker relative to his salary for a given season, but he was able to compensate for it on the defensive end with his prowess as an interior defender. Analyzing the traditional counting stats above will help lay the groundwork for future modeling efforts aimed at understanding player evaluation. Additional metrics relevant to player evaluation – such as age, position, and selected advanced statistics – are presented below and offer insights comparable to those of the previously discussed statistics.

Figure 6a: Average Age for each NBA Team by Season (2011-2025) vs Win Percentage – Teams: ATL to DAL

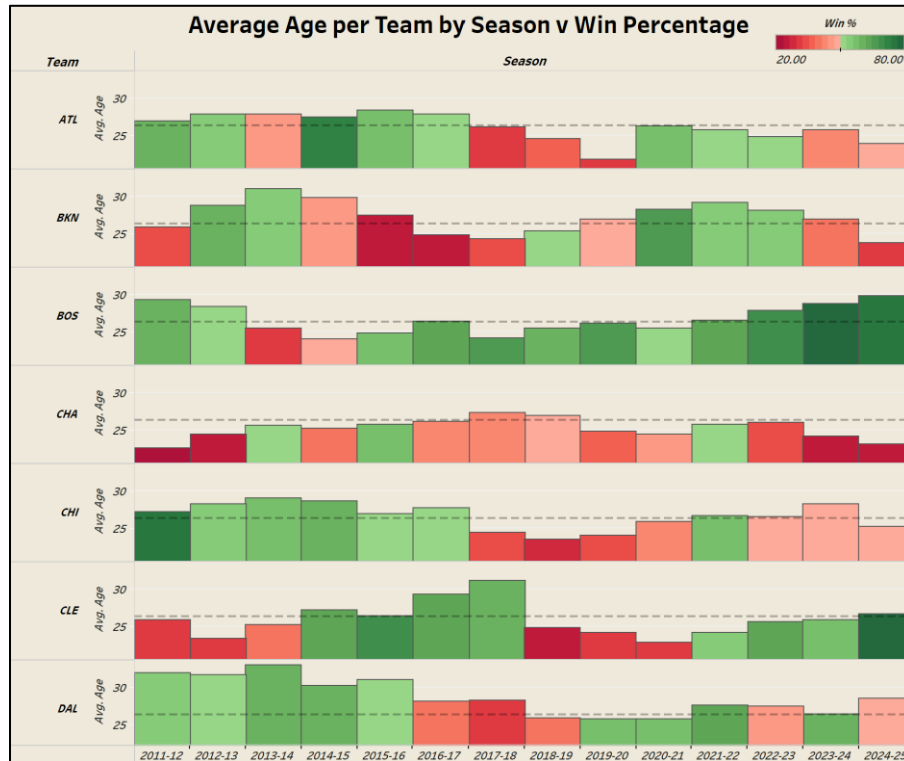


Figure 6b: Average Age for each NBA Team by Season (2011-2025) vs Win Percentage – Teams: DEN to LAL

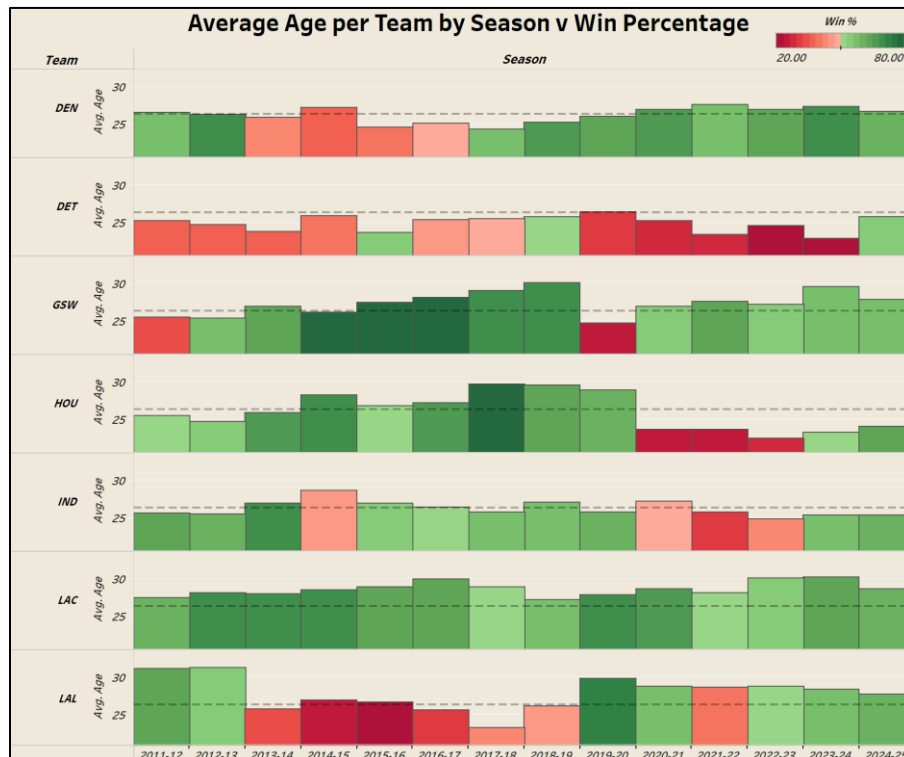


Figure 6c: Average Age for each NBA Team by Season (2011-2025) vs Win Percentage – Teams: MEM to OKC

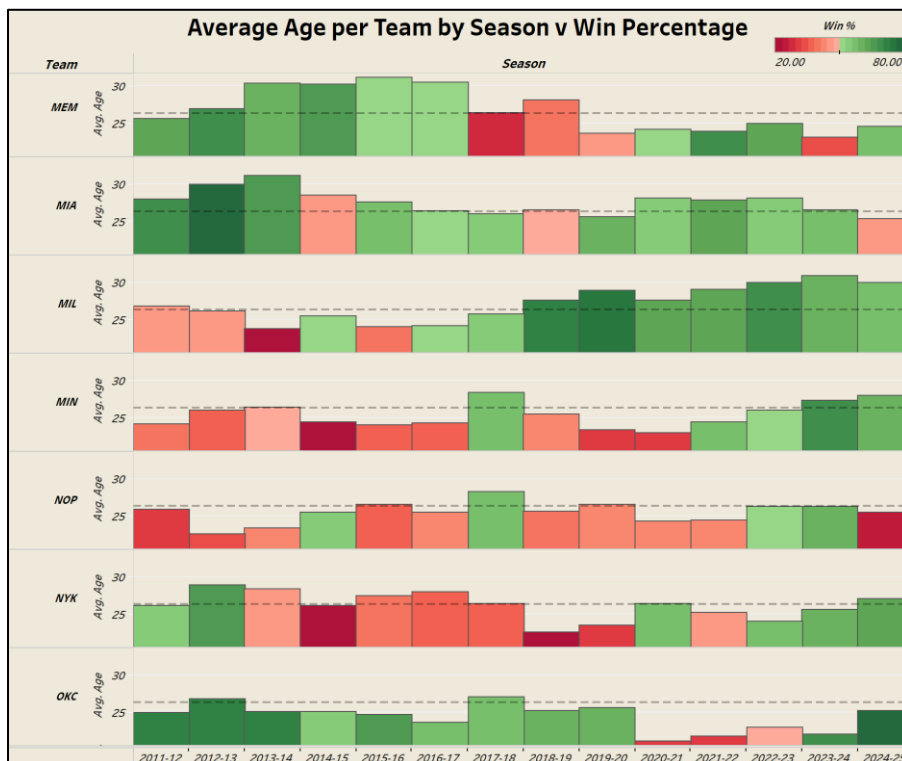


Figure 6d: Average Age for each NBA Team by Season (2011-2025) vs Win Percentage – Teams: ORL to TOR

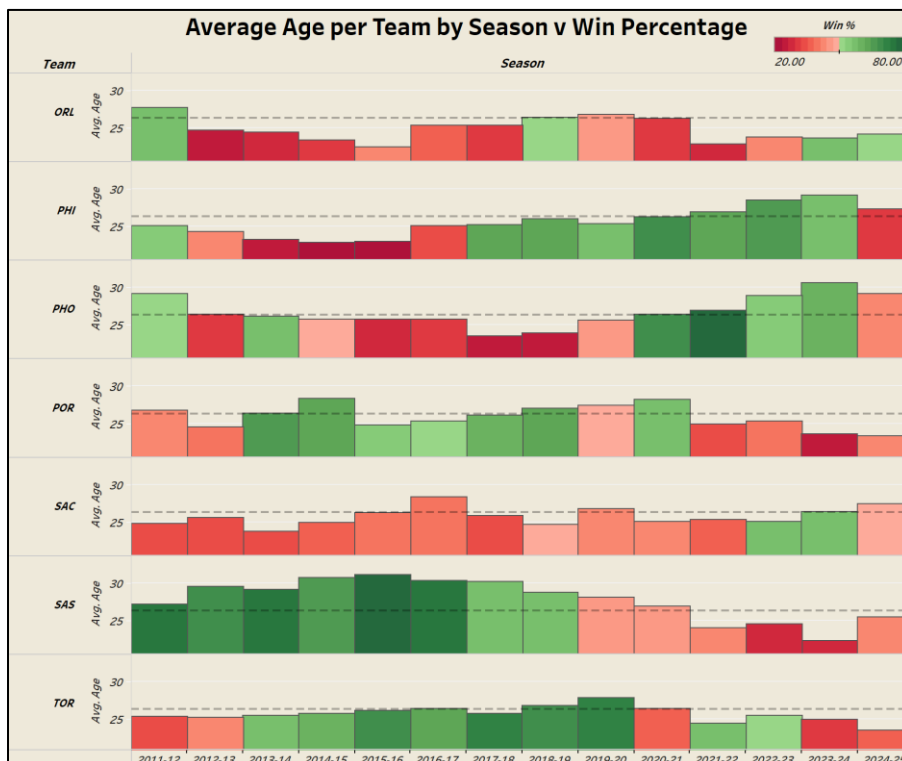
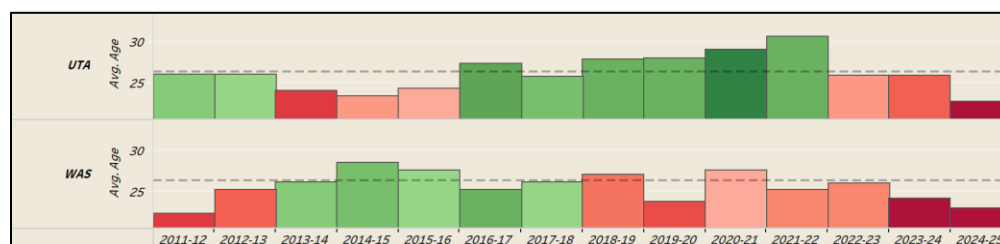


Figure 6e: Average Age for each NBA Team by Season (2011-2025) vs Win Percentage – Teams: UTA and WAS



As shown in Figures 6a-e, the average age on a roster, specifically for rotational players, between 2011 and 2025 is about 26.4 years old. As noted in the summary statistics section, the average age aligns with the fact that NBA players typically reach peak performance in their late 20s. Although all teams are included in the visualization, the following section will only focus on a select few – specifically the Clippers, Thunder, Kings, Pistons, and Spurs – for the sake of a more concise discussion.

Both the Los Angeles Clippers and Oklahoma City Thunder have sustained success over the past 14 seasons, though through very different team-building approaches. The Clippers' rotation has been older than the league average every season since 2011, reflecting the front office's preference for acquiring established veterans rather than committing to a rebuild. This is most evident in the franchise-defining trade for Paul George, which allowed the team to remain competitive during what could have been a transition period. Remarkably, the Clippers are the only team in the 14-year sample to post a win percentage above 50% every season, though this consistency hasn't translated into deep playoff success – they've reached the Western Conference Finals just once and have never made the NBA Finals in that span.

Coincidentally, the team on the other side of the Paul George trade – the Oklahoma City Thunder – has also been successful across this period, except for a few rebuilding years after the departures of Paul George and Russell Westbrook. Unlike the Clippers, OKC has leaned heavily on the draft, securing franchise talents like Kevin Durant, Russell Westbrook, Serge Ibaka, and, more recently, Chet Holmgren and Jalen Williams. Their ability to develop young stars on rookie-scale contracts has allowed them to spend elsewhere, as seen in the 2024 offseason when they signed Isaiah Hartenstein to a 3-year, \$87 million deal. While both strategies have kept these franchises competitive, the Thunder have enjoyed more playoff success historically and appear better positioned moving forward into the second half of the 2020s.

Now, the Sacramento Kings and Detroit Pistons have both struggled to find sustained success over the past 14 seasons, though their paths have differed significantly. The Kings have posted a winning record in only two seasons during this span, despite rarely fielding a particularly young core – only once since 2011 has their rotation's average age dipped below 24.75 years. While the franchise has attempted to build through the draft in recent years, many of its top young talents on rookie-scale deals – including DeMarcus Cousins, De'Aaron Fox, and Tyrese Haliburton – were either traded or failed to lead the team to consistent playoff appearances. Most recently, the Kings missed the postseason despite having their second-oldest rotation in the sample, suggesting the team may be entering another transitional phase.

In contrast, the Pistons have embraced youth much more aggressively. Detroit has had just two winning seasons since 2011, yet six of those years featured a rotation with an average age under 24.75 years. Notably, three of those youthful seasons occurred since 2021, each ending with a win percentage below 30% – a level of futility Sacramento never reached during this period. The Pistons’ front office made a clear commitment to rebuilding at the start of the 2020s, and while that has led to significant short-term losses, it now appears the foundation may be in place for long-term success. Heading into 2025–26, Detroit could be poised to turn the corner.

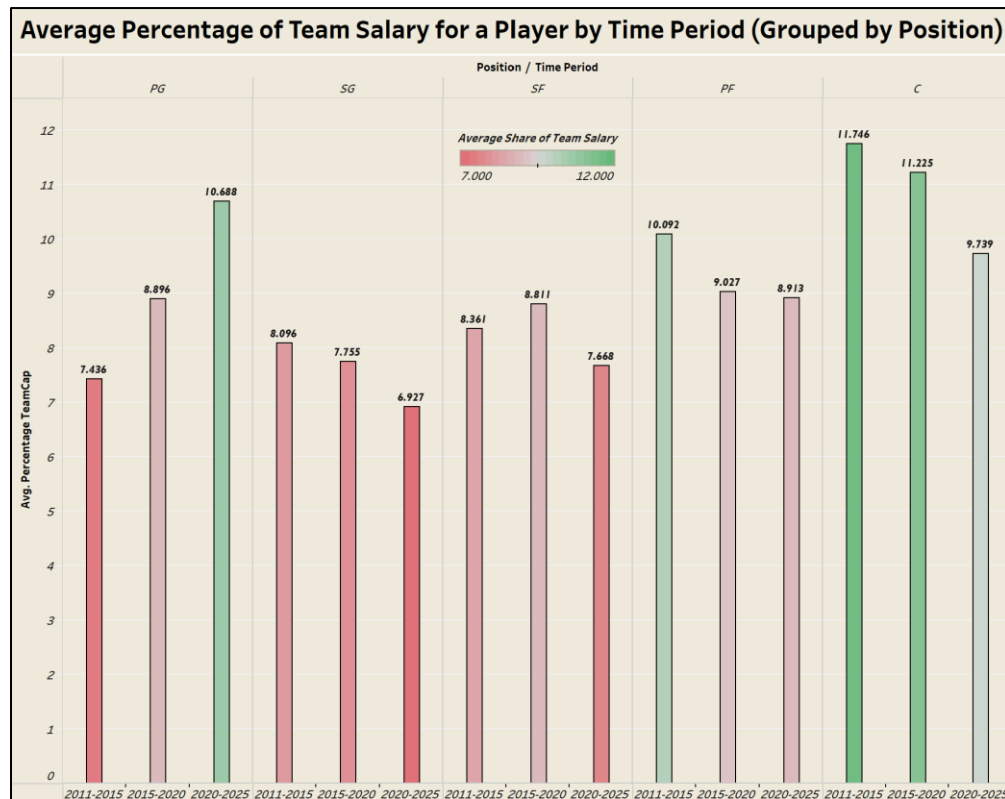
The contrasting trajectories of these franchises raise a compelling question: Which team appears better positioned for future success based on their strategy? Sacramento has seen limited success without fully embracing a youth-driven rebuild, while Detroit endured deeper lows with a younger roster, but may now be better positioned for sustained improvement.

The final team worth highlighting from Figures 6a–e is the San Antonio Spurs – a franchise that has experienced both prolonged success and extended struggles over the past 14 seasons. During much of the 2010s, the Spurs were defined by the presence of Tim Duncan and Kawhi Leonard. Between the 2011–12 season and Leonard’s trade to the Raptors in the 2018 offseason, San Antonio posted an average win percentage of 71.15% and fielded a veteran-heavy rotation with an average age of 29.81 years. During this period, the Spurs reached the NBA Finals twice and won the championship in 2014. However, their dominance quickly faded after Leonard's departure, resulting in just one winning season over the next seven years. By the 2021–22 season, it became evident that San Antonio was adopting a rebuild strategy focused on developing a young core, even if it meant enduring several losing seasons, to lay the foundation for long-term success. This approach mirrors that of the early-2020s Thunder and was similarly used by teams like the 2013–2016 76ers, 2017–2019 Suns, 2018–2021 Cavaliers, and 2019–2021 Timberwolves.

A common thread among these teams was their ability to eventually trade for, sign, or re-sign high-level players with large contracts – such as Joel Embiid, Jimmy Butler, and Tobias Harris (76ers); Devin Booker and Chris Paul (Suns); Donovan Mitchell and Jarrett Allen (Cavaliers); and Karl-Anthony Towns and Rudy Gobert (Timberwolves) – thanks to successful drafting that produced key contributors still on rookie-scale deals, including Ben Simmons (76ers); Mikal Bridges and Deandre Ayton (Suns); Evan Mobley and Darius Garland (Cavaliers); and Anthony Edwards (Timberwolves).

San Antonio appears to be following a similar path. With the recent addition of De’Aaron Fox via trade during the 2024–25 season, while Victor Wembanyama and Stephon Castle remain on rookie-scale contracts, the Spurs may be poised to replicate this roster-building formula and return to sustained success.

Figure 7: Average Percentage of Team Salary for a Player by Time Period, Grouped by Position



As illustrated in Figure 7, the market for NBA players by position has evolved, reflecting changes in offensive and defensive strategies, with the rise of three-point shooting and faster pace of play being particularly influential factors. For example, between 2011 and 2015, the average share of a team's total salary allocated to a rotational point guard was approximately 7.44%, but from 2020 to 2025, this share had risen to nearly 10.69%. In contrast, the salary share for a rotational center, which stood at 11.75% from 2011 to 2015, has decreased to about 9.74% in the more recent period. These shifts highlight the ongoing changes in team roster construction and the evolving nature of player evaluation. As teams continue to adjust their lineups, these trends are crucial for understanding how players may be valued in future free agency periods, emphasizing the need for updated charts that could be further refined on a season-by-season basis as new contracts are negotiated.

Figure 8: Win Shares per 48 Minutes & Player Efficiency Rating vs Percentage of Team Salary Taken Up by a Given Player

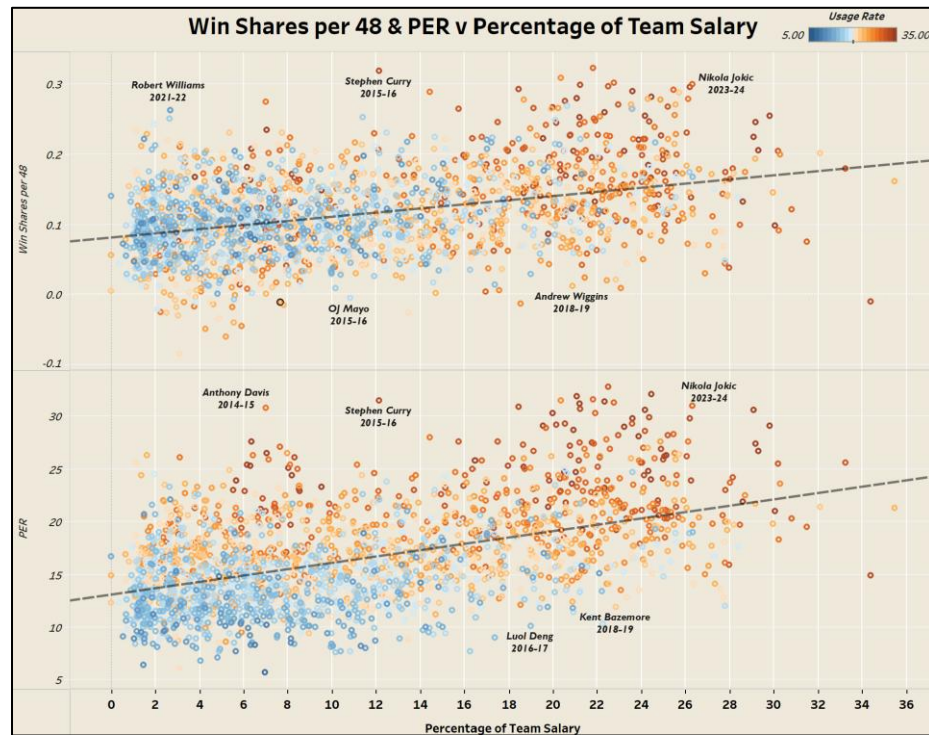
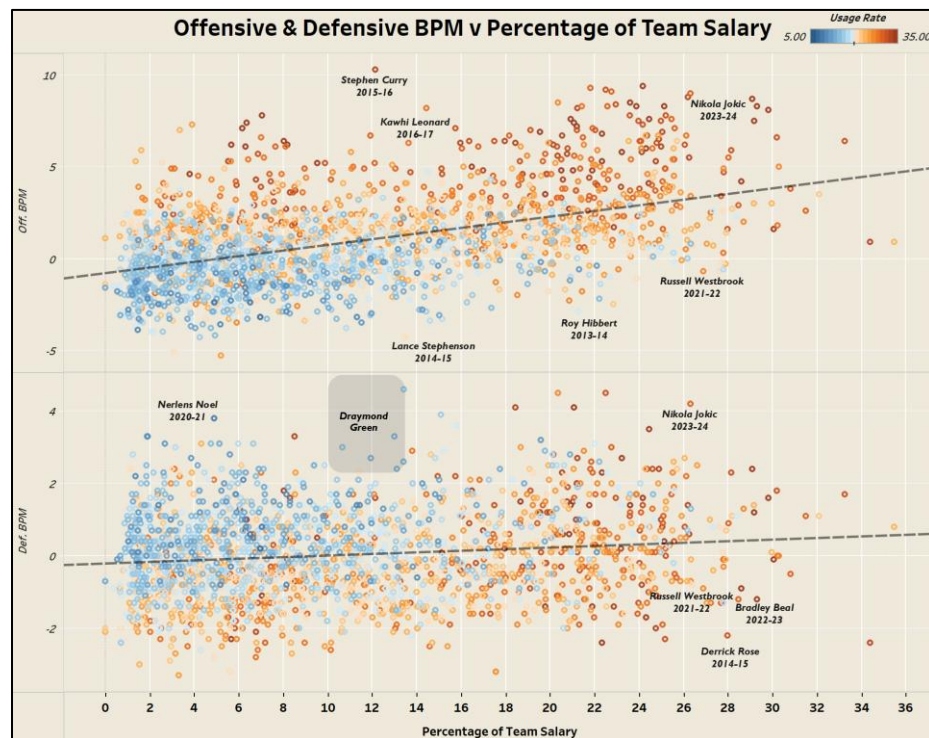


Figure 9: Offensive and Defensive Box Plus Minus vs Percentage of Team Salary Taken Up by a Given Player



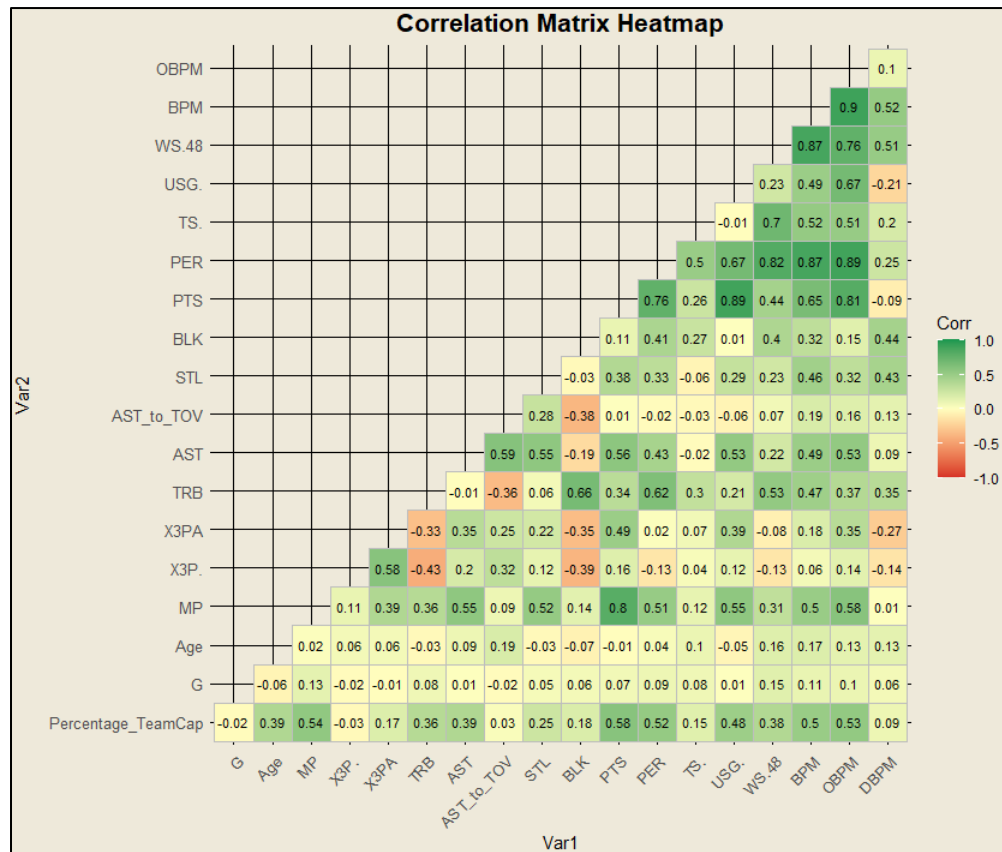
To further explore the relationship between player performance and salary allocation, Figures 8 and 9 illustrate the relationship between various advanced statistics – such as Win Shares per 48 Minutes (WS/48), Player Efficiency Rating (PER), Offensive Box Plus-Minus (OBPM), and Defensive Box Plus-Minus (DBPM) – and the share of team salary allocated to a player. Each figure also incorporates Usage Rate, which highlights players who can produce efficient statistics without necessarily dominating the ball. This trait is often associated with strong defenders or big men. Notable examples in these figures include Robert Williams, depicted in the top left of Figure 8 (WS/48), who was playing in the final season under a rookie-scale contract. His approximately 2% share of the Boston Celtics' total salary allowed the team to allocate sufficient funds to its star players, contributing to their success in reaching the NBA Finals during the 2021-22 season. Similarly, Draymond Green's elite defensive capabilities, highlighted on the DBPM graph in Figure 9, combined with his relatively modest salary, facilitated the Golden State Warriors' sustained success, including multiple NBA Finals appearances in the 2010s and early 2020s.

Examining the strength of the relationships between these metrics and player salary, it is evident that PER demonstrates a stronger correlation with player salary than WS/48. This difference can be attributed to PER's focus on traditional counting statistics, which tends to reward volume, while WS/48, being a rate stat, emphasizes efficiency. This suggests that teams may value players who contribute in volume, although context is crucial. A player's efficiency can fluctuate depending on their role and volume of play, meaning that players with lower volume may inherently exhibit higher efficiency.

In Figure 9, we further explore which advanced metric more accurately predicts player salary as a share of team salary. Both OBPM and DBPM exhibit positive relationships with salary, yet OBPM demonstrates a stronger correlation. It is important to note that OBPM primarily reflects individual offensive performance, whereas DBPM is more indicative of a player's impact on team defense, with some individual defensive contributions. The figure also highlights examples of players who either underperform or exceed expectations relative to their salary, such as Lance Stephenson, who fell below expectations, and Kawhi Leonard, who exceeded them.

While NBA front offices may not significantly, or even remotely, rely on these all-encompassing advanced metrics for player evaluation, these visualizations provide valuable insights that can inform models for assessing player performance and salary correspondence.

Figure 10: Correlation Matrix for Prospective Variables in a Player Evaluation Model



The correlation matrix in Figure 10 summarizes all metrics presented in prior charts and measures their correlation with one another and, more importantly, with the *Percentage_TeamCap* variable, which represents the share of team salary a player occupies. As acknowledged in the previous section concerning advanced statistics, the all-encompassing metrics such as PER, WS/48, and OBPM show high correlation with traditional volume metrics like Points per Game and Minutes per Game. This observation remains important as the analysis progresses toward multi-linear regression modeling to better understand player evaluation. The variables with the highest correlations with share of team salary are Points per Game, Minutes per Game, and OBPM, while those with the weakest correlations include Games Played, 3-Point Percentage, and Assist to Turnover Ratio. These results are generally intuitive: variables with stronger correlations reflect volume-based statistics, while those with weaker correlations either relate to health, which is somewhat controlled for in this dataset by including only players who appeared in at least 41 games, or to efficiency. In many cases, players who command a large share of their team's total salary continue to generate significant output through volume rather than efficiency.

Figure 11: Win Percentage vs Net Salary Spending Percentage for each NBA Team by Season (2011-2025)

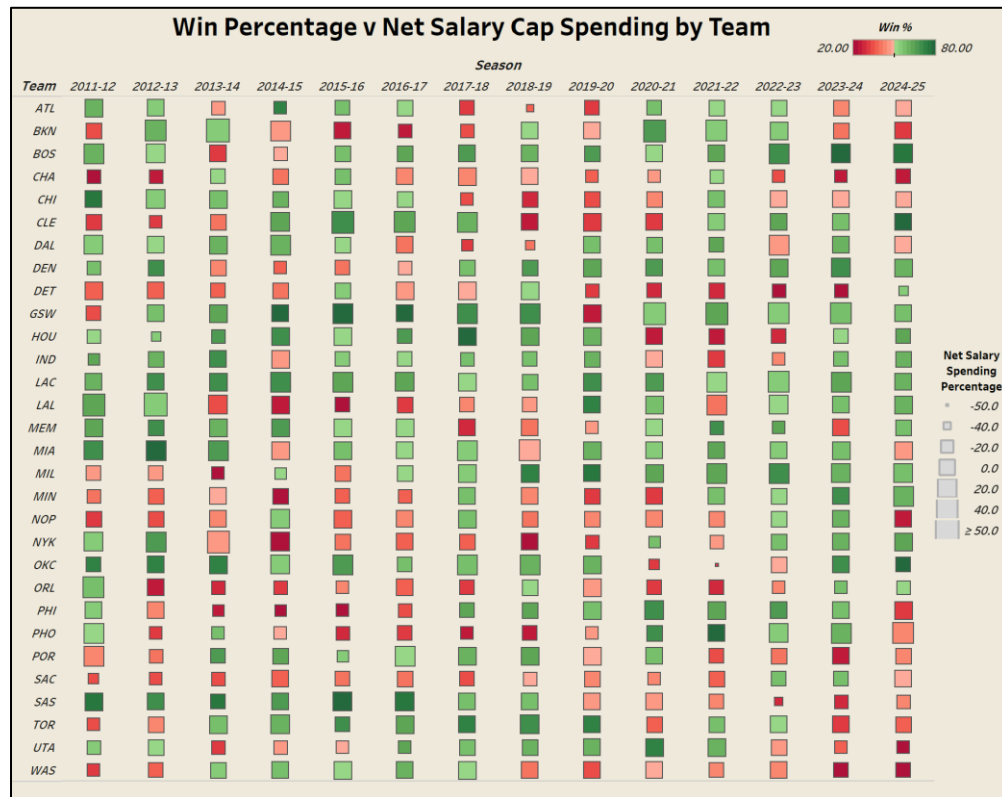


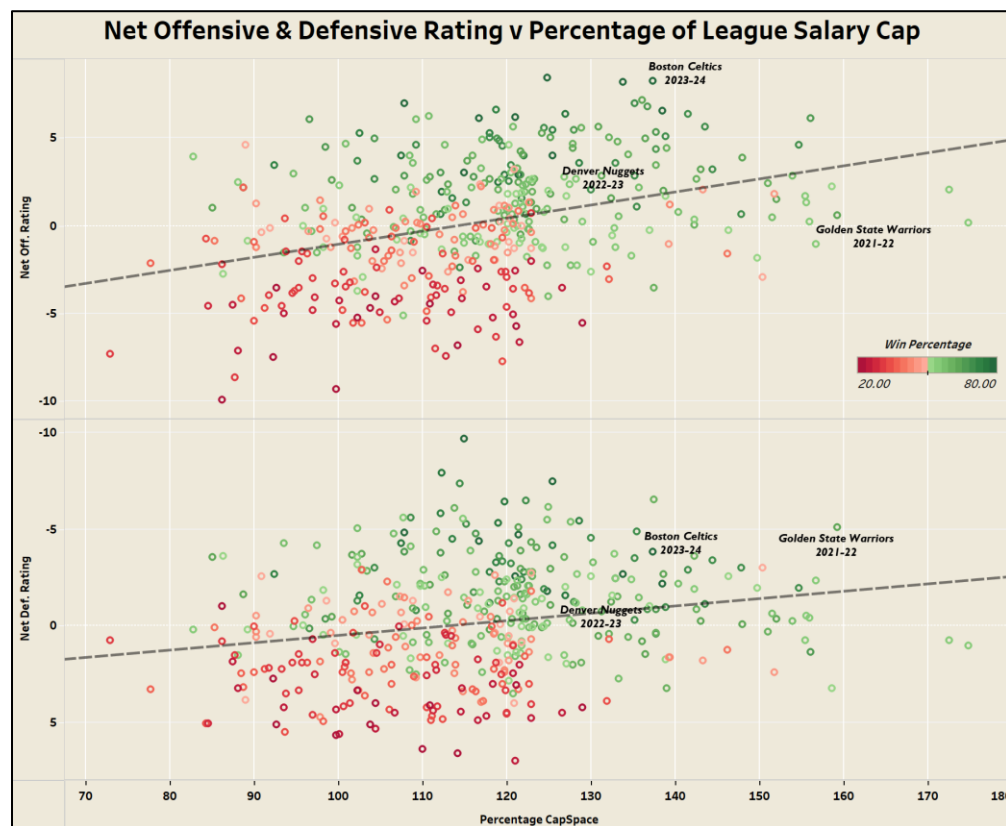
Figure 11 illustrates the relationship between net salary cap spending by a given team and its win percentage. The net salary metric represents the percentage of salary a team spends relative to the league average salary expenditure in a given season. For example, if Team A spends 125% of the league salary cap while the league average is 120%, the team's net spending would be 5%. This chart provides insight into how teams diverge in their willingness to spend and how such financial decisions correlate with on-court success in the same season. Similar to the previously analyzed team age chart, which helped uncover strategies such as leveraging rookie-scale contracts, Figure 11 helps identify additional team-building philosophies. For instance, franchises like the Oklahoma City Thunder in the early 2020s and the Phoenix Suns in the mid-to-late 2010s spent considerably less than league averages and endured losing seasons, with the goal of achieving sustained success in future years.

However, unlike Figures 6a–e, which emphasized maximizing value through the draft and efficient rookie contracts, Figure 11 introduces the alternative strategy of minimizing spending during rebuilding phases. This approach avoids investing heavily in free agents for short-term gains and instead focuses on meeting the league's minimum salary threshold with smaller, more tradable contracts that can yield younger talent or draft capital. The Detroit Pistons provide a recent example, entering the 2024–25 season with the league's lowest team salary yet achieving a winning record. This success stems from refraining from overpaying free agents in recent years, although contracts such as Tobias Harris's warrant closer examination when analyzing spending efficiency. Looking ahead, Cade Cunningham's five-year extension begins in 2025–26 at approximately \$38.7 million, increasing by 8% annually through 2029–30. If the

Pistons maintain this spending discipline, they may pursue a marquee free agent or star via trade in the upcoming offseason, although specific opportunities will remain unclear until the offseason begins.

It is also important to consider how the 2023 CBA's introduction of the second apron – discussed previously in this paper – reshapes future spending strategies. As shown in Table 5, the second apron threshold stands at approximately 134% of the league salary cap. Exceeding this level carries substantial penalties, limiting a team's flexibility to sign players, make trades, or engage in long-term planning. Consequently, team-building models like the Golden State Warriors' in the mid-2010s to early 2020s are no longer sustainable under the new rules. Attempts to replicate such approaches are unlikely to yield long-term success, evident by the Phoenix Suns' current roster and constrained outlook.

Figure 12: Net Offensive and Defensive Rating vs. Team Salary Cap Usage as a Percentage of League Salary Cap for a Given Season



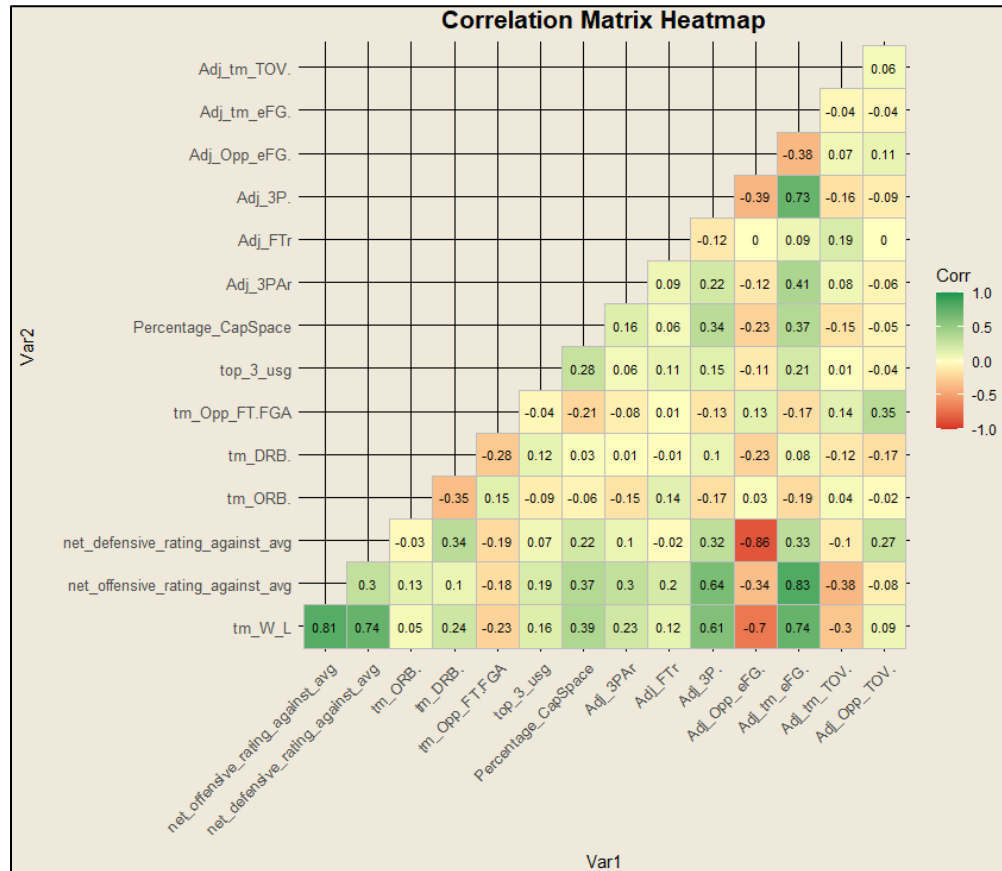
Taking a closer look at the relationship between Offensive Rating and Defensive Rating in relation to the percentage of the league salary cap spent by teams in a given season, Figure 12 suggests that having a stronger offense relative to the league average may be more highly valued than having a superior defense when it comes to team spending. It is important to note that Net Offensive Rating, Net Defensive Rating, and Percentage of Cap Space (defined as the portion of the league salary cap spent on player salaries) are all normalized for the specific season they occurred in. This adjustment ensures that a top-10 offense in 2024, for instance, is interpreted in the proper context relative to earlier seasons – such as 2012 – where that same offensive rating might have ranked in the top three, if not higher.

Although the last three NBA champions are labeled in Figure 12 and could be examined as case studies to understand whether higher spending leads to elite success, such analysis is not recommended due to the extremely limited sample size, which is unlikely to yield meaningful conclusions for team-building strategies. Nonetheless, all three championship teams spent more than 130% of the league's salary cap. With the 2023 Collective Bargaining Agreement (CBA) introducing a second apron projected at 134% of the cap, as shown in Table 5, it will be important to monitor how these high-spending thresholds correlate with team success in future seasons.

Table 5: League Salary Cap and other Cap Thresholds (2011-2035)*

[illegible]

Figure 13: Correlation Matrix for Prospective Variables in a Team Style Evaluation Model



Finally, before transitioning to the models that evaluate roster construction strategies, Figure 14 presents a correlation matrix containing all variables that may be used in the analysis. A preliminary review reveals that the variables most strongly correlated with Win Percentage (*tm_W_L*) are the Net Offensive and Net Defensive Rating metrics previously discussed. Following closely are Adjusted Effective Field Goal Percentage, Adjusted Opponent Effective Field Goal Percentage, and Adjusted 3-Point Percentage. It is important to note that any variable labeled with 'Adj' has been normalized relative to the league average for the season in which the team competed. This adjustment accounts for shifts in league-wide playstyle, such as the increasing emphasis on three-point shooting, since the 2011–12 season. Much like the earlier matrix shown in Figure 10, this visualization establishes a baseline for interpreting and contextualizing the results of the forthcoming models.

Modeling and Methodology

Table 6: OLS Regression Results for Player Contract Evaluation

Dependent variable: Percentage TeamCap			
PosPF	-1.574*** (0.563)	BLK	0.444* (0.264)
PosPG	-4.892*** (0.738)	PTS	0.571*** (0.033)
PosSF	-2.226*** (0.644)	TS.	-0.104*** (0.022)
PosSG	-3.358*** (0.676)	PosPF:Era2015-2020	-0.026 (0.726)
Era2015-2020	-0.479 (0.551)	PosPG:Era2015-2020	1.260* (0.718)
Era2020-2025	-1.920*** (0.575)	PosSF:Era2015-2020	0.858 (0.732)
MP	0.154*** (0.033)	PosSG:Era2015-2020	0.396 (0.708)
Age	0.678*** (0.022)	PosPF:Era2020-2025	0.657 (0.725)
X3PA	-0.080 (0.064)	PosPG:Era2020-2025	3.159*** (0.737)
X3P.	-0.026*** (0.010)	PosSF:Era2020-2025	1.259* (0.732)
TRB	0.147** (0.071)	PosSG:Era2020-2025	1.728** (0.708)
AST	0.535*** (0.092)	Constant	-13.660*** (1.469)
AST_to_TOV	-0.026 (0.180)	Observations	3,176
STL	0.139 (0.301)	R ²	0.561
		Adjusted R ²	0.558
		Residual Std. Error	4.812 (df = 3150)
		F Statistic	161.162*** (df = 25; 3150)
		Note:	*p<0.1; **p<0.05; ***p<0.01

In analyzing the first of three models, focused specifically on player valuation, most variables from the original correlation matrix in Figure 10 were initially included. However, after several iterations (one of which is presented in the Appendix, Table 9), composite advanced statistics such as PER, WS/48, OBPM, and DBPM were removed. This decision was driven by the recognition that front offices are more likely to base salary decisions on a player's visible, in-game impact rather than complex advanced metrics. Furthermore, these variables explained little additional variation in the dependent variable (share of team salary), and their inclusion risked overfitting and overcomplicating the model. Usage Rate was also excluded due to its limited contribution to model performance and potential multicollinearity with Minutes Per Game and Points Per Game, as all three metrics reflect a player's offensive volume.

Categorical variables for position and era were included to account for how positional roles and evolving league playstyles influence player compensation. This approach was motivated by the insights from Figure 7, which examined average salary share by position across

different eras. Additionally, interaction terms between position and era were incorporated to capture the shifting valuation of specific positions over time, particularly the increasing importance of point guards from 2011 to 2025.

The final model, seen in Table 6, explains approximately 56.1% of the variation in a player's share of team salary and is statistically significant overall. Significant predictors at the 5% level included: all positional categories, the most recent era, Minutes Per Game, Age, 3-Point Percentage, Rebounds Per Game, Assists Per Game, Points Per Game, True Shooting Percentage, and the interaction terms between guard positions and the most recent era.

Key findings reveal that age and offensive production metrics – notably Points and Assists per Game – have the strongest influence on a player's share of team salary. Specifically, for each additional point or assist per game, a player is expected to earn 0.571% and 0.535% more of their team's salary cap, respectively, holding all else constant. Similarly, each additional year of age is associated with a 0.678% increase in salary share, all else equal. Most notably, a point guard in the 2020–2025 era is expected to earn 3.159% more of a team's salary cap than a center in the 2011–2015 era, highlighting how positional value and team-building strategies have evolved under changing league conditions. Additionally, it is important to note that the positive coefficient on age may be partially driven by the structural constraints of the NBA's salary system, specifically, the presence of rookie-scale contracts. Younger players who outperform expectations are often unable to immediately capitalize on their market value due to contract limitations, whereas older players are typically on their second or third contracts, which better reflect their performance. Therefore, the age coefficient should be interpreted with caution, as it may capture contract phase effects as much as actual age-based valuation

Table 7: OLS Regression Results for the Relationship between Team Playstyle and Team Success

Dependent variable:	
	tm_W_L
tm_ORB.	1.276*** (0.089)
tm_DRB.	1.216*** (0.115)
Adj_3PAr	0.001 (0.058)
Adj_FTr	0.559*** (0.093)
Adj_3P.	0.325 (0.205)
Adj_Opp_eFG.	-4.525*** (0.170)
Adj_tm_eFG.	4.498*** (0.197)
Adj_tm_TOV.	-4.283*** (0.262)
Adj_Opp_TOV.	3.104*** (0.219)
Percentage_CapSpace	0.042*** (0.015)
Constant	-77.841*** (9.942)
Observations	420
R ²	0.913
Adjusted R ²	0.911
Residual Std. Error	4.450 (df = 409)
F Statistic	431.444*** (df = 10; 409)
Note:	* p<0.1; ** p<0.05; *** p<0.01

Shifting focus to team-level performance, Table 7 presents the regression model evaluating the impact of team playstyles on win percentage. As noted previously, all on-court metrics, except for Offensive and Defensive Rebound Percentage, have been adjusted relative to league averages for their respective seasons, ensuring comparability across years. For brevity, adjusted metrics are referred to without the "Adj" prefix here.

The model is statistically significant and explains 91.3% of the variance in team win percentage, indicating a strong fit. Key statistically significant predictors at the 5% level include: Offensive Rebound Percentage (ORB%), Defensive Rebound Percentage (DRB%), Free Throw Rate (FTR), Opponent Effective Field Goal Percentage (Opp eFG%), Effective Field Goal Percentage (eFG%), Turnover Percentage (TOV%), Opponent Turnover Percentage (Opp TOV%), and team salary cap spending as a percentage of the league cap.

Notably, a one-unit increase in Turnover Percentage is associated with a 4.3 percentage point decrease in win percentage, holding all else constant. In contrast, a one-unit increase in Opponent Turnover Percentage yields only a 3.1 percentage point increase in win percentage. This asymmetry suggests that the negative impact of committing turnovers outweighs the

benefits of forcing them, highlighting the strategic value of ball security over defensive aggression.

Regarding team spending, the coefficient on salary cap spending indicates that each 1% increase in spending relative to the league cap is associated with a 0.042% increase in win percentage, holding all else constant. For context, a team increasing spending from the cap level to just below the second tax apron (a 33% increase) would be expected to improve its win percentage by approximately 1.386%.

It is noteworthy that 3-Point Attempt Rate and 3-Point Percentage are not statistically significant in the model, indicating that no definitive conclusions can be drawn regarding the impact of three-point shooting on team success on a year-to-year basis. However, the results do suggest that teams with more aggressive playstyles – such as those that draw frequent fouls (as reflected in higher Free Throw Rates) and crash the offensive boards – are generally expected to achieve higher win percentages, as indicated by the statistically significant coefficients in the above output.

Lastly, the coefficients on eFG% and Opponent eFG% are similar in magnitude but opposite in sign, implying that improvements in shooting efficiency and defensive shot suppression contribute nearly equally to team success. This motivates further analysis in the final model to determine whether offense or defense, relative to league averages, is more predictive of winning.

Table 8: OLS Regression Results for the Relationship between Offensive Rating/Defensive Rating and Team Success

	Dependent variable:
	tm_W_L
net_offensive_rating_against_avg	2.872*** (0.064)
net_defensive_rating_against_avg	2.896*** (0.071)
Percentage_CapSpace	0.036*** (0.013)
Constant	44.986*** (1.495)
Observations	420
R ²	0.933
Adjusted R ²	0.932
Residual Std. Error	3.882 (df = 416)
F Statistic	1,930.354*** (df = 3; 416)
Note:	*p<0.1; **p<0.05; ***p<0.01

The model in Table 8 includes only three variables but is able to explain 93.3% of the variance in win percentage. Both net offensive and net defensive rating variables were discussed previously, specifically with Figure 12, and the salary cap percentage variable was addressed in the last section. All three variables, as well as the overall model, are statistically significant at the 1% level, allowing for confident interpretation of the results.

First, the coefficient for the salary cap variable is slightly smaller than in the second model – 0.036 compared to 0.042 – indicating a modestly reduced impact on win percentage in this specification. As for the key question raised in the previous section – is offense or defense more influential in determining win percentage relative to league averages? – the results indicate that net defensive rating has a slightly greater effect. Specifically, a one-unit increase in net defensive rating (relative to league average) is associated with an expected 2.896% increase in win percentage, compared to a 2.872% increase for net offensive rating, holding all else constant. Thus, defensive performance relative to the league appears to be slightly more impactful than offensive performance when it comes to predicting team success.

Future Considerations

Future extensions of this research could more directly examine the impact of the 2023 Collective Bargaining Agreement's introduction of a second tax apron, specifically how it influences team decisions and behavior in subsequent seasons, potentially including an examination of profit-maximizing and win-maximizing teams. Furthermore, future research could benefit from analyzing whether successful team playstyles or strategies differ between the

regular season and the playoffs. Given the heightened intensity, slower pace, and more targeted defensive schemes in postseason basketball, identifying which regular-season strengths translate to playoff success may offer further insight into optimal team construction. Due to time constraints, further work is also planned to analyze the roster construction of current teams and to forecast how the 2025 offseason may unfold for certain franchises, using the foundational insights presented in this paper. Additionally, incorporating more advanced and robust modeling techniques could yield more precise conclusions and deepen the understanding of team-building strategies in the NBA. As additional seasons and off-seasons take place under the 2023 CBA framework, it will be important to revisit and update these models to reflect evolving league trends and strategies. Finally, a project adding to this one, but instead focusing primarily on rookie-scale deals or any other type of contract structures and their impact on success could be insightful.

Conclusion

This paper aimed to examine which roster construction strategies most effectively position NBA teams for sustained success within the constraints of the modern salary cap and collective bargaining agreements. Three regression models were developed: the first focused on individual player evaluation based on performance metrics, position, age, and era; the second and third models explored team-level strategies, including playstyle preferences and the comparative value of offensive versus defensive strength.

The first model revealed that age and offensive production – particularly Points and Assists per Game – are the most influential factors in determining a player's share of team salary. Furthermore, results suggested a growing valuation of point guards in the modern NBA, especially since the early 2010s, highlighting a broader league-wide shift in team-building priorities that favor offensive-minded guards.

The subsequent models offered insight into team playstyle optimization. Limiting turnovers emerged as more critical to success than forcing them, and aggressive offensive strategies – such as drawing fouls and dominating the offensive and defensive glass – were positively associated with win percentage. Interestingly, no statistically significant relationship was found between three-point shooting (both in volume and accuracy) and team success on a year-to-year basis.

Finally, the third model directly addressed the offense-versus-defense debate. It concluded that, while both are important, having a stronger defense relative to the league average contributes slightly more to winning than having a stronger offense.

In summary, this research identifies the key factors in both player valuation and roster construction that contribute to success under the NBA's evolving economic framework. While offensive skill and guard play are highly valued in individual contracts, team success is more strongly tied to disciplined, defense-oriented strategies.

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Appendix

Table 8: OLS Regression Results for Player Contract Evaluation (including composite advanced stats)

Dependent variable:			
	Percentage_TeamCap		
PosPF	-1.725*** (0.566)	BLK	0.759** (0.359)
PosPG	-4.917*** (0.740)	PTS	0.620*** (0.099)
PosSF	-2.303*** (0.655)	PER	-0.288 (0.177)
PosSG	-3.358*** (0.680)	TS.	-0.177*** (0.043)
Era2015-2020	-0.232 (0.562)	USG.	0.001 (0.085)
Era2020-2025	-1.427** (0.618)	WS.48	0.339** (0.143)
MP	0.111* (0.059)	OBPM	0.233 (0.189)
Age	0.672*** (0.022)	DBPM	-0.238 (0.175)
X3PA	-0.181* (0.092)	PosPF:Era2015-2020	-0.002 (0.727)
X3P.	-0.025** (0.010)	PosPG:Era2015-2020	1.212* (0.719)
TRB	0.164* (0.090)	PosSF:Era2015-2020	0.797 (0.734)
AST	0.619*** (0.103)	PosSG:Era2015-2020	0.344 (0.709)
AST_to_TOV	-0.262 (0.220)		
STL	0.554 (0.503)		

PosPF:Era2020-2025	0.666 (0.726)
PosPG:Era2020-2025	3.038*** (0.740)
PosSF:Era2020-2025	1.202 (0.735)
PosSG:Era2020-2025	1.641** (0.710)
Constant	-6.539* (3.374)
Observations	3,176
R ²	0.563
Adjusted R ²	0.559
Residual Std. Error	4.808 (df = 3145)
F Statistic	134.921*** (df = 30; 3145)
Note:	* p<0.1; ** p<0.05; *** p<0.01